

Introduction

Baroid Core Homework 6

Assessment Feedback

Congratulations, you passed.

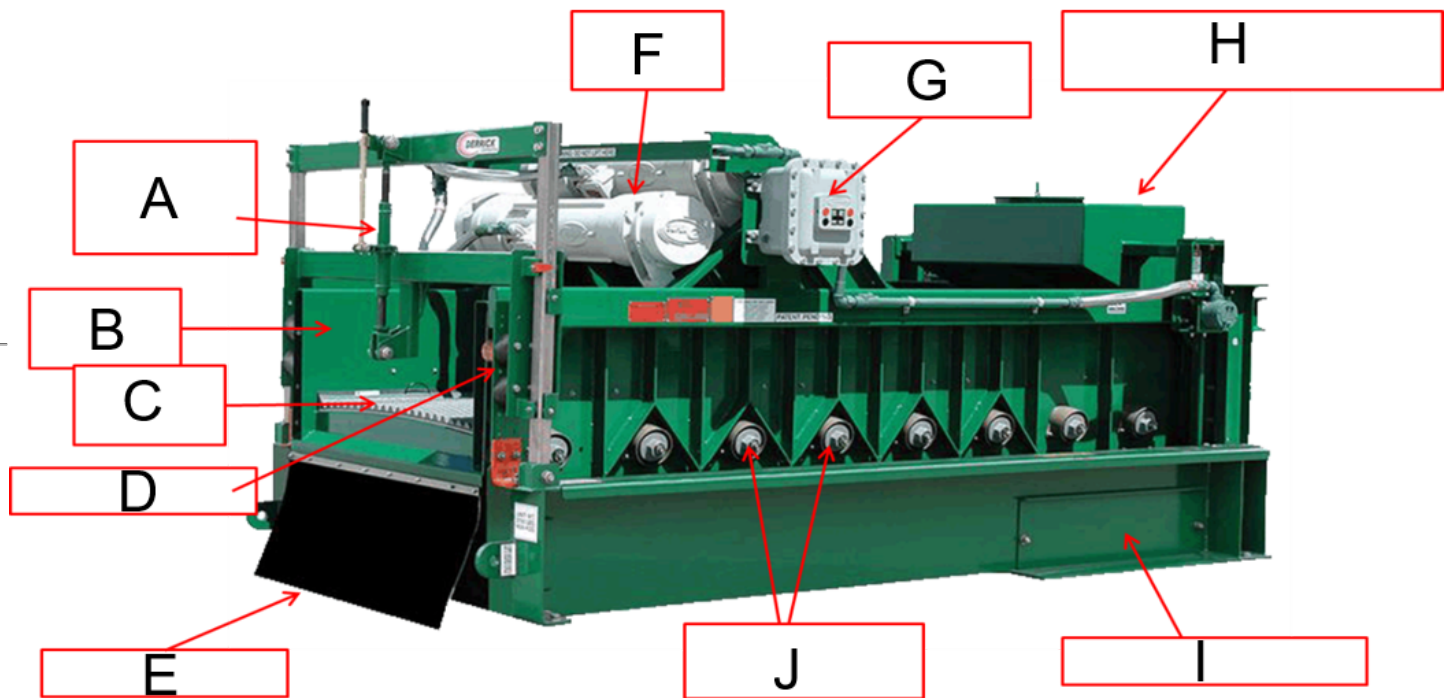
Total score: 75 out of 75, 100%

Question Feedback

Question

1 of 40

Label the parts of an FLC-504 shaker below using the following diagram:



- Screen Discard
- Feed Hopper/Back tank/ Possum belly
- Control Box
- Float mounts
- Screen Tension
- Vibrator Motors
- Jack Assembly (AWD)
- Vibrating Basket
- Screen Deck
- Screen Under Flow

10 out of 10

Question

2 of 40

Which piece of solids control equipment is capable of removing 60-80% of the total generated drilled solids from the drilling fluid if operated efficiently?

- ☐ Flo-Divider
- ☐ Mud Cleaner
- ☒ Shale Shaker
- ☐ Vertical Dryer
- ☐ Centrifuge

1 out of 1

Question

3 of 40

What is the most important aspect of a shale shaker in regards to the size of solids removed?

- ☒ Screen selection
- ☐ Basket angle
- ☐ Motor motion
- ☐ G-force
- ☐ RPM of motor

1 out of 1

Question

4 of 40

Select the three main types of motion shale shakers utilize:

- ☐ Oscillating
- ☒ Linear
- ☒ Circular
- ☐ Rhombic
- ☐ Oval
- ☒ Elliptical

3 out of 3

Question

5 of 40

What shaker motion is the best choice for finer screening and increased basket angles?

- ☐ Elliptical
- ☐ Centrifugal
- ☐ Oscillating
- ☐ Circular
- ☒ Linear
- ☐ Rhombic

1 out of 1

Question

6 of 40

What is the name of the API standard established in 2004 to allow individuals to easily compare screens?

- ☒ API-RP-13C
- ☐ API-RP-13B-1
- ☐ API-RP-13B-2
- ☐ API-RP-13A

1 out of 1

Question

7 of 40

The established API standard for screen classification was created to standardize the following:

- ☐ Tension
- ☒ Non-blanked area
- ☒ Labeling
- ☐ Conveyance rate
- ☐ Aspect ratio
- ☒ Separation size
- ☐ Wire durability
- ☒ Conductance

4 out of 4

Question

8 of 40

Screen **Conductance** is defined as “The movement of drilling fluid and classified solid particles through the shaker screens.”

1 out of 1

Question

9 of 40

Screen **Conveyance** is defined as “The movement of unclassified solid particles off the shaker screens”

1 out of 1

Question

10 of 40

Shaker performance can **GENERALLY** be defined by the following parameters

- ☐ G-force
- ☒ Dryness of solids discharge
- ☒ Solids removal
- ☒ Processing Capacity
- ☐ Basket weight
- ☐ Number of screen panels

3 out of 3

Question

11 of 40

What does screen mesh mean?

- ☐ Material of the screens construction
- ☒ Number of openings per linear inch for square mesh screens
- ☐ Dimensions of the area the screens cover inside the Shaker
- ☐ Thickness in 1/32nd of inches

1 out of 1

Question

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What is the cut point potential of shaker screens that have satisfied the present API classification standards? —

- ☐ D₅₀
- ☐ D₁₆
- ☒ D₁₀₀
- ☐ D₈₄

1 out of 1

Question

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What unit of measurement is displayed on the API label of the screen to indicate screen conductance? —

- ☐ Nanodarcy
- ☐ Millidarcy (mD)
- ☒ Darcy (kD/mm)
- ☐ None of the above

1 out of 1

Question

14 of 40

Provide the contributing features that enable BaraMesh screens to consistently outperform the market competitors in regards to shaker *screen durability* and *particle blinding prevention*:

A) BaraMesh feature providing increased Screen Durability:

Increased wire diameter ▼

B) BaraMesh feature providing improved Particle Blinding resistance:

Altered aspect Ratio from 1:1 (square) to 1:1.6 (rectangular) ▼

2 out of 2

Question

15 of 40

Provide the most significant benefits of using “corrugated” screens over “conventional” screens: —

- ☐ Reduced particle blinding
- ☒ Ability to screen finer earlier
- ☒ Increased screen conductance
- ☐ Reduced screen cost per unit
- ☐ Increased screen durability
- ☒ Increased non-blanked screening area

3 out of 3

Question

16 of 40

When using a four screen shaker, what is the “fluid end point” rule of thumb during drilling operations? —

- ☐ ¼ of the screens covered
- ☐ As close to end of last screen as possible
- ☒ ¾ of the screens covered
- ☐ 50% of the screens covered
- ☐ 80% of the screens covered
- ☐ None of the above

1 out of 1

Question

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What are the two types of screens typically used in oil and gas drilling operations in reference to “*tension*”?

- ☒ Field-tensioned screen
- ☐ Flexible frame screens
- ☒ Pre-tensioned (rigid frame) screens
- ☐ Corrugated triangular-tensioned screens

2 out of 2

Question

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What are the two types of screens typically used in oil and gas drilling operations in reference to “*dimension*”?

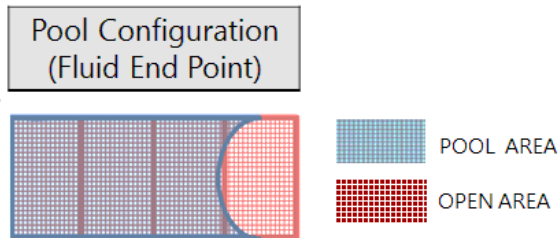
- ☐ Field-tensioned screens
- ☒ Corrugated (pyramid/3D) screens
- ☒ Conventional (flat/2D) screens
- ☐ Advanced (trapezoid/3D) screens

2 out of 2

Question

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Using the diagram below, enter what adjustments, if any, need to made to the basket angle, screens, or flow rate to achieve the proper fluid end point



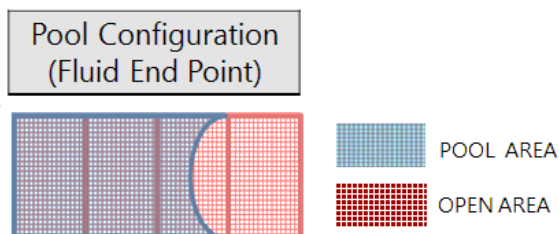
- ☐ Increase basket angle
- ☐ Decrease basket angle
- ☐ Screen down (coarser)
- ☐ Screen up (finer)
- ☐ Increase flow rate
- ☐ Decrease flow rate
- ☒ None of the above

3 out of 3

Question

20 of 40

Using the diagram below, enter what adjustments, if any, need to made to the basket angle, screens, or flow rate to achieve the proper fluid end point



- ☐ Increase basket angle

- ☒ Decrease basket angle
- ☐ Screen down (coarser)
- ☒ Screen up (finer)
- ☒ Increase flow rate
- ☐ Decrease flow rate
- ☐ None of the above

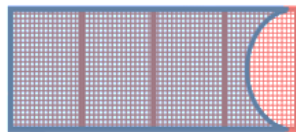
3 out of 3

Question

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Using the diagram below, enter what adjustments, if any, need to be made to the basket angle, screens, or flow rate to achieve the proper fluid end point

Pool Configuration (Fluid End Point)



 POOL AREA
 OPEN AREA

- ☒ Increase basket angle
- ☐ Decrease basket angle
- ☒ Screen down (coarser)
- ☐ Screen up (finer)
- ☐ Increase flow rate
- ☒ Decrease flow rate
- ☐ None of the above

3 out of 3

Question

22 of 40

Non-blanked area of a screen refers to which of the following:

- ☐ The total area of the shaker basket
- ☐ The total area in which fluid covers the screen (fluid end-point)
- ☐ The total area of the panel (length x width)
- ☒ The total area that fluid can pass through
- ☐ The number of openings per square linear inch

1 out of 1

Question

23 of 40

List two negative consequences of the solids conveyance rate being too slow?

- ☒ Excessive bridging and derogation to solid particles causing screen blinding
- ☐ Solid particles form clumps and take longer to separate
- ☐ Solid particles become hotter
- ☒ Increased wear and tear (damage) to screen

2 out of 2

Question

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One negative consequences of the solids conveyance rate being too fast, is less dewatering/drying time causing increased surface fluid loss and excessive waste

- ☒ True
☐ False

1 out of 1

Question

25 of 40

Which of the following are **NOT** a suggested operating guideline for shaker screens?

- ☐ Run finest mesh possible while covering 3/4 of the screen area.
☒ Use finer screens towards the back of the shaker and coarser for the drying panel
☒ Open header box bypass valve when sending hole cleaning sweeps to avoid losing mud
☐ Run Screen frame as level as possible to reduce screen wear.
☐ Ensure screens are properly mounted on a clean screen frame.

2 out of 2

Question

26 of 40

What is the G-force of a shaker with a stroke length of 0.173 inches and a motor RPM of 1750RPM?
Use the formula below to calculate (use 1 decimal for answer)

$$G\text{-force} = \frac{\text{Stroke}_{\text{length}} \times (RPM^2)}{70414}$$

G-force is (round using 1 decimal)

3 out of 3

Question

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A is a device / mechanism utilized to provide the length of movement (in inches) the shaker motors are providing.

1 out of 1

Question

28 of 40

What are the common types of couplings used on pumps in the oilfield industry? (select all that apply)

- ☐ Flexible rubber
☒ Dodge
☒ Love joy
☐ Chain
☐ Flexible glass

2 out of 2

Question

29 of 40

Which type of pump will degrade solids the fastest?

- ☐ Progressive Cavity
☒ Centrifugal
☐ Rotary lobe
☐ Diaphragm

- ☐ Triplex pump

1 out of 1

Question

30 of 40

What type of pump should **NOT** be used to feed centrifuges?

- ☒ Centrifugal
- ☐ Progressive cavity
- ☐ Lobe

1 out of 1

Question

31 of 40

Choose the **two (2)** pump characteristics associated with centrifugal pumps?

- ☐ Proportional Flow
- ☒ Cavitation Prone
- ☐ Reversible
- ☒ Solid Degrading
- ☐ Gentle “conveying” action

2 out of 2

Question

32 of 40

The **Impeller**  is the component of the centrifugal pump positioned on a rotating axis (shaft) imparting the mechanical force on the drilling fluid needed to transport the fluid to its receiving point

1 out of 1

Question

33 of 40

A Progressive Cavity (PC) pump utilizes what components working in tandem to transfer drilling mud?

- ☐ Impeller / Volute
- ☒ Rotor / Stator
- ☐ Gears / Lobes
- ☐ Cylinder / Piston Rod

1 out of 1

Question

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Which factor causes the most wear in PC pumps?

- ☐ Pressure
- ☒ Speed
- ☐ Temperature
- ☐ Chemicals
- ☐ All of the above

1 out of 1

Question

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Why is the anti-spin bolt on centrifugal pump impellers important? —

- ☐ Makes the pump last longer
- ☐ Makes the pump more efficient
- ☒ Saves you from breaking the Volute/Casing/Snail/Impeller
- ☐ It has no particular purpose

1 out of 1

Question

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Do the rotating lobes touch each other, in a rotary lobe pump? —

- ☐ Yes
- ☒ No

1 out of 1

Question

37 of 40

Should you ever run a pump dry? —

- ☐ Yes
- ☒ No

1 out of 1

Question

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What is the most common pump found on drilling rigs surface pit systems? —

- ☐ Rotary Lobe
- ☐ Progressive Cavity
- ☐ Diaphragm
- ☒ Centrifugal
- ☐ None of the above

1 out of 1

Question

39 of 40

Choose the most critical characteristics of a positive displacement pump in regards to its use feeding a decanting centrifuge: —

- ☒ Proportional flow rate to speed (RPM)
- ☐ High shearing velocity
- ☐ Solid degrading
- ☒ Variable speed
- ☒ Gentle conveyance

3 out of 3

Question

40 of 40

To ensure hydrocyclones are operating properly and efficiently they require

- ☐ 75 psi at the feed inlet of the hydrocyclone
- ☐ 75 feet of head at the discharge (outlet) of the centrifugal pump
- ☒ 75 feet of head at the feed inlet of the hydrocyclone
- ☐ 75 psi at the inlet of the centrifugal pump

1 out of 1